Solution Requirements

Date	02 October 2026
Team ID	SWTID-2026-1529
Project Name	EduGenie: Google gemini powered learning assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Core AI Processing	System must generate MCQs, summaries, and learning roadmaps.	
1	Authentication	*[Describe how users will securely log in or verify their identity]*	
2	User Authentication	Users must be able to sign up, log in, and view study history.	
3	Data Export	Users can download quizzes in JSON and notes in Markdown format.	
2	Authorization levels	*[Define the user roles and what access/permissions each role has]*	
3	External interfaces	*[Detail any third-party APIs, external databases, or*	

		services the system must connect to]	
4	Transactions processing	*[Explain how data exchanges, payments, or core system transactions will be handled]*	
5	Reporting	*[Describe the types of reports, dashboards, or data exports the system must generate]*	
6	Business rules, etc.	*[List specific logic, constraints, or workflows unique to the business process]*	
7	Compliance to laws or regulations	*[Specify any legal, industry, or data privacy standards the system must adhere to (e.g., GDPR, HIPAA)]*	
8	*Other*	*[Add any additional functional requirements here]*	

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	API latency for Q&A should be minimal.	< 2 seconds response time
2	Reliability	JSON outputs from Gemini API must be strictly structured.	100% JSON schema validation rate

1	Performance & Speed	*[Describe response times, processing speed, etc.]*	*(e.g., Page loads in < 2 seconds)*
2	Scalability	*[Describe how the system handles growth in users or data]*	*(e.g., Supports up to 10,000 concurrent users)*
3	Scalability	Backend must handle concurrent student requests	Support 50+ concurrent requests
4	Usability	Frontend must be responsive and accessible	Lighthouse Score > 90
3	Security & Data Privacy	*[Detail data protection, encryption, and vulnerability management]*	*(e.g., End-to-end AES-256 encryption)*
4	Reliability & Availability	*[Specify uptime requirements and fault tolerance]*	*(e.g., 99.9% uptime per month)*
5	Usability & Accessibility	*[Describe user experience standards and accessibility compliance]*	*(e.g., Complies with WCAG 2.1 AA standards)*
6	*Other*	*[Add any additional non-functional requirements (e.g., Maintainability, Portability)]*	